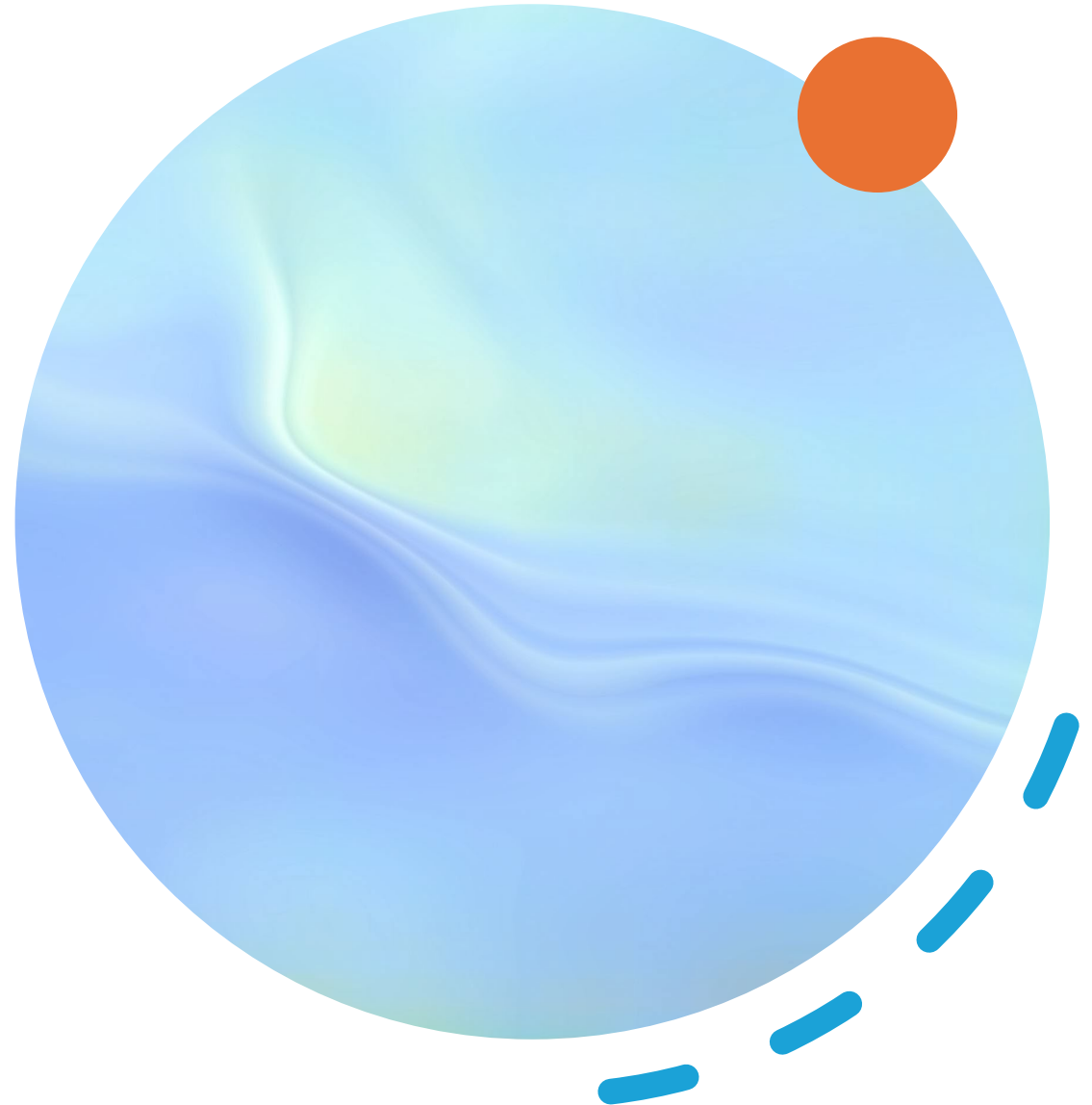
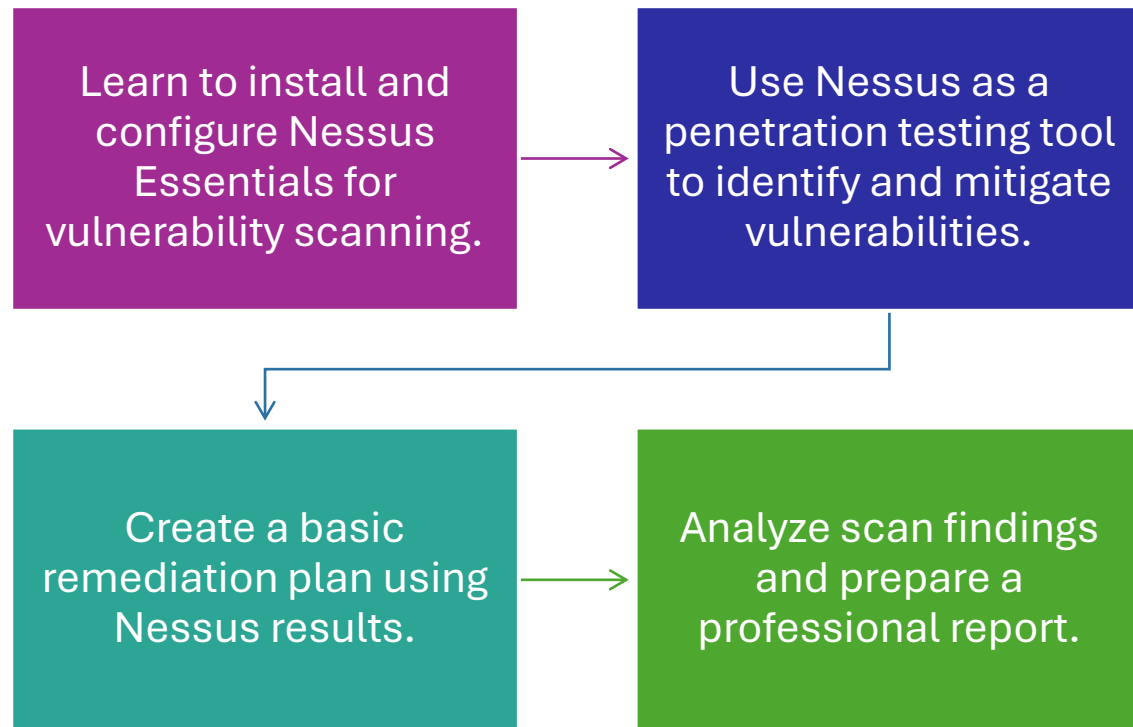




Lab 2: Network Scanning and Remediation Plan Creation Using Nessus

CSIT302 Cybersecurity | University of Wollongong Dubai

Objectives



Introduction to Nessus

What is Nessus?

- A vulnerability scanner developed by Tenable.
- Used to detect vulnerabilities, misconfigurations, and compliance issues.
- Trusted by security professionals for penetration testing and vulnerability assessments.

Key Features:

- Intuitive web interface.
- Pre-configured scanning templates.
- Real-time detection and risk analysis.
- Customizable reporting options.

Why Nessus?

- Helps identify weaknesses before attackers exploit them.
- Provides actionable insights for improving security posture.

Prerequisites

Nessus Essentials installed
(download from Tenable's
official website).

Login credentials for Nessus
web portal ready.

Target system or a network with
permission to scan.

Activity 1 - Conducting a Basic Network Scan

Configuring the Scan:

- Open Nessus and navigate to the "New Scan" option.
- Select the Basic Network Scan template.
- Enter a scan name (e.g., "Lab 2 Network Scan") and specify a target IP range.
- Save the scan configuration.

Running the Scan:

- Start the scan and monitor its progress in real-time.
- Wait for scan completion.

Analyzing Results:

- Open the scan report.
- Identify vulnerabilities categorized as Critical, High, Medium, and Low.
- Examine detailed descriptions and recommendations for each vulnerability.

Activity 2 - Using Nessus as a Penetration Testing Tool

Simulate Exploitable Scenarios:

- Select a vulnerability from the scan results.
- Assess the potential impact of the vulnerability.
- Document the steps attackers might take to exploit it.

Create a Security Policy:

- Focus on critical vulnerabilities first.
- Outline required patches, configuration changes, and best practices to prevent exploitation.

Activity 3 - Reporting and Documentation

Prepare a Lab Report:

- Include details of the scan configuration and target environment.
- Summarize findings, focusing on critical vulnerabilities.
- Outline the proposed remediation plan with mitigation steps.
- Add lessons learned and next steps for improving network security.

Deliverables for Lab Report

Detailed analysis of at least two vulnerabilities (focus on critical vulnerability, if any) and their mitigation steps.



Screenshots of Nessus scan results and follow-up scans to verify mitigation success.



Lessons learned and one proposition to improve the scanned network security.

Key Takeaways

Nessus provides a powerful platform for identifying and mitigating vulnerabilities.

Regular scanning and proactive policies enhance an organization's security posture.

Understanding scan reports is essential for effective penetration testing.

Discussion Questions

Why is regular network scanning important?

How can Nessus improve an organization's security?

What challenges could arise when using Nessus?